

Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The purpose of this report is to analyze the Geldium delinquency prediction dataset, assess data quality, identify missing values and anomalies, and determine early indicators of customer delinquency risk before predictive modeling.

2. Dataset Overview

The dataset contains 500 customer records and 19 variables consisting of numerical and categorical features related to customer demographics, financial status, credit behavior, and delinquency outcome.

Key dataset attributes:

- Number of records: 500
- Number of variables: 19
- Data types: Numerical and categorical
- Target variable: Delinquent_Account

Initial observations:

- No duplicate records were identified during the initial review.
- A few variables contain missing values.
- The target variable is imbalanced (approximately 420 non-delinquent and 80 delinquent customers).

3. Missing Data Analysis

Missing Data Issue	Handling Method	Justification
Income (39 missing)	Median Imputation	Median is robust to outliers and preserves realistic income distribution.
Loan_Balance (29 missing)	Median Imputation	Prevents data loss while maintaining central tendency.
Credit_Score (2 missing)	Mean/Median Imputation	Very few missing values; simple imputation has minimal impact.

Initial data quality observations:

- Missing values are limited to a few important financial variables.
- No evidence of excessive missing data requiring column removal.

- Median imputation is appropriate for skewed financial variables.
- After imputation, the dataset will be suitable for predictive modeling.

4. Key Findings and Risk Indicators

High-risk indicators:

- High Credit Utilization – Customers using a high percentage of available credit are more likely to become delinquent.
- Low Credit Score – Indicates weaker repayment history and higher default risk.
- High Loan Balance – Large outstanding balances increase repayment burden.
- Previous Late Payments – Strong indicator of future delinquency.
- Low Income with High Debt – May reduce repayment capacity.

Additional insights:

- The dataset contains an imbalanced target variable, so class balancing techniques may improve model performance.
- Missing financial values should be imputed before training.
- Credit behaviour variables are expected to contribute more strongly than demographic variables.

5. AI & GenAI Usage

Generative AI was used to:

- Summarize the dataset.
- Recommend strategies for handling missing values.
- Identify likely delinquency risk factors.
- Draft the EDA report.

Example prompts:

- Summarize key patterns, outliers and missing values in this dataset.
- Suggest an imputation strategy for missing income values.
- Identify the top three variables likely to predict delinquency.

6. Conclusion & Next Steps

The dataset is generally clean and suitable for predictive analytics after addressing a small number of missing values. Credit utilization, credit score, loan balance, and payment history appear to be the most influential indicators of delinquency risk. Recommended next steps include completing data imputation, performing feature engineering, balancing the target classes if necessary, and developing a classification model such as Logistic Regression or

Decision Tree for delinquency prediction.